

1.

i

$$16_{10} \rightarrow 010000_2 \rightarrow \begin{array}{r} 101111 \\ + \\ 110000 \\ \hline \end{array}$$

2's complement
11110000

$$29_{10} \rightarrow 011101_2 \rightarrow \begin{array}{r} 100010 \\ + \\ 1 \\ \hline 100011 \end{array}$$

2's complement
100011

16+29

$$\begin{array}{r} 010000 \\ + 011101 \\ \hline 101101_2 \end{array}$$

6-bit 2's complement has a valid range of -32 to 31
= 45 $\notin$ [-32, 31]

NO OVERFLOW [DOES NOT REQUIRE EXTRA BITS]

16-29

$$\begin{array}{r} 010000 \\ + 100011 \\ \hline 110011 \end{array}$$

Since $16-29 = -13$ which is between -32 and 31,
No overflow

ii

26^m Binary

$$-26_{10} \rightarrow 011010_2 \rightarrow \begin{array}{r} 100101 \\ + \\ 1 \\ \hline 100110 \end{array} \rightarrow \text{26 m Binary}$$

$$19_{10} \rightarrow 010011_2 \rightarrow \begin{array}{r} 101100 \\ + \\ 1 \\ \hline 101101 \end{array}$$

2's complement is

-26+19

$$\begin{array}{r} 100110 \\ + 010011 \\ \hline 111001 \end{array}$$

this value is -7, since $-32 \leq -7 \leq 31$ ✓

No overflow

-26-19

$$\begin{array}{r} 100110 \\ + 101101 \\ \hline 101001 \end{array}$$

this value is < -32 , OVERFLOW